TRT PROTOCOLS | BLOOD WORK GUIDE | HORMONE OPTIMIZATION | SIDE EFFECT MANAGEMENT

TESTOSTERONE REPLACEMENT THERAPY (TRT)

COMPLETE OPTIMIZATION GUIDE

Protocols · Dosing · Blood Work · Aromatase Management · Thyroid · Sexual Health

Evidence-based hormone optimization for health, vitality & body composition

ELITE FITNESS GUIDE SERIES	Research-Based • Professional Edition • 2024

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■ 1. LOW TESTOSTERONE — SYMPTOMS, CAUSES & DIAGNOSIS

Low testosterone (hypogonadism) affects an estimated **2–4% of adult men**, with subclinical low-normal testosterone (300–450 ng/dL) affecting a much larger population. Symptoms often present years before lab values fall below clinical cutoffs, causing significant quality-of-life impairment.

> ■ *A major American urological study found that men with total testosterone <300 ng/dL have a 42% increased cardiovascular risk, significantly reduced bone density, and a 3× higher incidence of metabolic syndrome compared to men with optimal T levels (600–900 ng/dL).*

Symptoms of Low Testosterone:

Category	Symptoms	Severity
Physical	Decreased muscle mass, increased body fat (especially abdominal), reduced strength, fatigue even after adequate sleep, decreased body/facial hair	Often attributed to aging; frequently missed for years
Sexual	Reduced libido, erectile dysfunction, decreased morning erections, reduced ejaculate volume, infertility	Major quality of life impact; often first symptom noticed
Psychological	Depression, anxiety, irritability, reduced motivation, cognitive fog, poor concentration, low confidence	Often misdiagnosed as depression — antidepressants are NOT the solution if cause is hormonal
Metabolic	Insulin resistance, type 2 diabetes risk, dyslipidemia (high LDL, low HDL), visceral fat accumulation	Forms a negative feedback loop — fat cells convert T to estrogen, worsening T deficiency

■ 2. THE HYPOTHALAMIC-PITUITARY-TESTICULAR (HPT) AXIS

Understanding the HPT axis is essential for interpreting blood work, understanding why TRT suppresses natural production, and making informed decisions about HCG, AI, and PCT:

- **Hypothalamus** releases GnRH (Gonadotropin-Releasing Hormone) in pulses every 1–3 hours
- **Pituitary gland** responds to GnRH by releasing LH (Luteinizing Hormone) and FSH (Follicle-Stimulating Hormone)
- **LH** travels to Leydig cells in the testes → stimulates testosterone production
- **FSH** travels to Sertoli cells → stimulates sperm production (spermatogenesis)
- **Testosterone** feeds back to hypothalamus and pituitary → reduces GnRH, LH, FSH (negative feedback)
- **TRT effect:** Exogenous testosterone triggers strong negative feedback → LH and FSH fall to near zero → testes stop producing testosterone and sperm → testicular atrophy over months

This is why **HCG is critical** on TRT for men who want to maintain fertility or prevent complete testicular atrophy.

■ 3. TRT PROTOCOLS — TYPES, DOSES & ADMINISTRATION

Protocol	Dose	Frequency	Pros	Cons
Testosterone Enanthate (IM)	100–200 mg	Every 7–14 days	Inexpensive; familiar; stable	Peaks and troughs with 14-day schedule; requires injection
Testosterone Enanthate (Sub-Q)	80–150 mg	Every 5–7 days (split)	More stable levels; less SHBG elevation; smaller needle	Less conventional; requires more frequent injections
Testosterone Cypionate (IM)	100–200 mg	Every 7–14 days	Slightly longer half-life than Enanthate; widely available	Same as Enanthate
Testosterone Propionate	25–50 mg	Every 2–3 days	Very stable blood levels; fast adjustment	Daily or EOD injections; more painful
Testosterone Gel/Cream	50–100 mg topical	Daily application	Non-invasive; very stable levels	Transfer risk to women/children; variable absorption
Testosterone Pellets	Implanted pellets	Every 3–6 months	Zero administration burden	Surgical insertion; dose adjustment difficult

Optimal TRT Dose Principle: The goal is to bring total testosterone into the **upper-normal physiological range (700–1000 ng/dL)**, not simply above the clinical cutoff (300 ng/dL). Many men feel optimal at 900+ ng/dL while still remaining within a physiological range that minimizes cardiovascular and prostate risk.

■ 4. FREE VS TOTAL TESTOSTERONE — OPTIMIZATION

Total testosterone measures ALL testosterone in blood. However, only **2–4% of testosterone is 'free'** (unbound to proteins) and biologically active. The rest is bound to SHBG (Sex Hormone Binding Globulin) or albumin.

■ *Research demonstrates that free testosterone is a superior predictor of clinical outcomes (muscle mass, libido, energy) compared to total testosterone. A man with total T of 700 ng/dL but very high SHBG may have the same symptoms as a man with total T of 400 ng/dL and normal SHBG.*

How to Optimize Free Testosterone:

- **Reduce SHBG:** SHBG rises with age, insulin sensitivity, thyroid dysfunction, and low androgen levels. TRT itself reduces SHBG somewhat
- **Optimize thyroid:** Hypothyroidism dramatically increases SHBG — correcting T3/T4 liberates bound testosterone
- **Maintain healthy body weight:** Obesity increases SHBG and aromatase activity simultaneously
- **Adequate zinc:** Zinc inhibits SHBG synthesis in the liver — supplement 25–50 mg/day
- **Boron:** 10 mg/day boron has been shown to reduce SHBG by 9% and increase free T by 25% in a clinical study
- **Injection frequency:** More frequent TRT injections (2–3×/week) maintain more stable free T levels vs once-weekly peaks

■ 5. ESTROGEN MANAGEMENT ON TRT

Estrogen management on TRT requires a more nuanced approach than during cycle use. **Men on TRT need adequate estrogen** for cardiovascular health, bone density, cognitive function, and sexual desire. Aggressive AI use on TRT is one of the most common and harmful mistakes.

> ■■ **IMPORTANT:** Multiple studies show that men with very low estrogen on TRT have WORSE cardiovascular outcomes than men with optimally elevated estrogen. Do NOT attempt to suppress E2 to low levels — it increases atherosclerosis risk, reduces bone density, and causes sexual dysfunction.

TRT Estrogen Targets:

- Total T 700–1000 ng/dL with E2 30–50 pg/mL = optimal for most men (some feel best at E2 40–60 pg/mL)
- Symptoms matter more than numbers — adjust AI based on how you feel AND blood work together
- If AI is needed: Anastrozole 0.25 mg twice weekly maximum — start lower than AAS protocols
- Consider Aromasin over Arimidex for TRT — suicidal AI provides less fluctuation in E2 levels

■ 6. HCG ON TRT — FERTILITY & TESTICULAR HEALTH

- **Standard TRT dose:** 500 IU HCG 2× weekly alongside TRT injections
- **Fertility-focused:** 1000–2000 IU 3× weekly + consider FSH (Gonal-F or Follistim) if sperm count remains low
- **Testicular atrophy prevention:** 250 IU EOD (smaller, more frequent) maintains testicular size better than large infrequent doses
- **Mechanism:** HCG mimics LH — directly stimulates Leydig cells in testes to maintain function and size
- **HCG storage:** Reconstitute with bacteriostatic water; refrigerate; stable 60 days

■ 7–9. THYROID · DHT · BLOOD WORK GUIDE

Thyroid Optimization on TRT:

The thyroid-testosterone relationship is bidirectional — low T reduces T3/T4 conversion, and low thyroid further depresses testosterone. Testing TSH alone is insufficient; demand **full thyroid panel** (TSH, Free T3, Free T4, Reverse T3, TPO antibodies).

Test	Optimal Range	Action if Out of Range
TSH	0.5–2.0 mIU/L	TSH >2.5 with symptoms: consider thyroid support
Free T3	3.5–4.2 pg/mL	Low Free T3 with normal T4: impaired T4→T3 conversion; consider T3 supplementation
Free T4	1.2–1.7 ng/dL	Low T4: primary hypothyroidism; may require T4 (Synthroid) or combination T4+T3
Reverse T3	<15 ng/dL	High rT3 = T3 is being blocked; often caused by chronic stress or caloric restriction

DHT & Hair Loss Management:

- Testosterone converts to DHT via 5-alpha reductase enzyme — DHT is the primary cause of male pattern baldness (MPB) in genetically susceptible men
- 5-alpha reductase inhibitors (Finasteride 1mg, Dutasteride 0.5mg) reduce scalp DHT by 60–90%
- Caution: Finasteride can cause permanent sexual side effects ('Post-Finasteride Syndrome') in a small percentage of men — research thoroughly before use
- Alternative: Topical Finasteride (0.025–0.1% solution applied to scalp) — reduces scalp DHT with minimal systemic absorption

Complete TRT Blood Work Panel:

Frequency	Panel
Baseline (before TRT)	Total T, Free T, LH, FSH, E2, Prolactin, SHBG, DHT, CBC, CMP, Lipids, PSA, TSH, Free T3, Free T4, HbA1c, Ferritin, Vitamin D
3 months post-start	Total T, Free T, E2, SHBG, CBC (hematocrit), PSA, Lipids, CMP
Every 6 months (stable)	Total T, Free T, E2, SHBG, CBC, PSA, Lipids, CMP
Annually	Full panel same as baseline

❤ 10–12. CARDIOVASCULAR · SEXUAL HEALTH · LIFESTYLE

Cardiovascular Monitoring on TRT:

- TRT increases red blood cell production — check hematocrit every 3–6 months. If >52%: donate blood or reduce dose
- TRT may worsen dyslipidemia (especially HDL). Omega-3 4–6g/day is non-negotiable
- Blood pressure: Monitor monthly; target <130/80. Reduce sodium; exercise cardiovascular training
- Recent large studies (TRAVERSE trial, 2023) found TRT does NOT increase cardiovascular events in hypogonadal men with pre-existing cardiovascular disease

Optimizing Sexual Function on TRT:

- Libido usually improves within 3–6 weeks of TRT initiation
- If libido remains low despite optimal T: check E2 (too high or too low), prolactin (elevated = dopamine issue), thyroid (low T3), and zinc/vitamin D deficiency
- PT-141 (Bremelanotide): 1–2 mg SC 45–90 min before activity — acts centrally on desire circuits regardless of hormonal status
- Tadalafil (Cialis): 5 mg daily (low-dose) improves blood flow and may improve T response by increasing testicular blood circulation

Lifestyle Synergy with TRT: Resistance training 4×/week (amplifies TRT response by upregulating AR density). Sleep 7–9 hours (testosterone peaks during REM sleep — poor sleep negates TRT). Maintain body fat 10–18% (higher fat = more aromatization; lower fat = less conversion). Minimize alcohol (directly suppresses Leydig cell function).